PYTHON PHASE-1 MASTER QUESTION LIST

A. Basics, Numbers & Logic

1. Print your name, age, branch.
2. Take input and print uppercase.
3. Swap two numbers (with third variable).
4. Swap without third variable.
5. Swap without arithmetic operators.
6. Convert Celsius to Fahrenheit.
7. Convert km to miles.
8. Simple interest.
9. Compound interest.
10. Area of circle.
11. Area of triangle.
12. Convert seconds to hh:mm:ss.
13. Check positive/negative/zero.
14. Even or odd.
15. Maximum of 3 numbers.
16. Minimum of 3 numbers.
17. Factorial (loop).
18. Factorial (recursion).
19. Reverse number.
20. Palindrome number.
21. Count digits.
22. Sum of digits.
23. Armstrong number.
24. Perfect number.

25. Strong number.

26. Automorphic number.

27. Harshad number.

28. Sunny number.

29. Digital root.

30. Sum of proper divisors.

31. Find missing number (1 to n).

32. GCD using loop.

33. GCD using Euclidean method.

34. LCM.

35. Count trailing zeros in factorial.

36. Decimal to binary (manual).

37. Binary to decimal (manual).

38. Check power of 2.

39. Count set bits.

40. Find nth prime.

41. Next palindrome number.

42. Basic calculator using if-elif.

B. Strings (From Basic to Advanced)

43. Reverse string.

44. Palindrome string.

45. Count vowels.

46. Count consonants.

47. Count spaces.

48. Remove spaces.
49. Convert case without built-in.
50. Length without len().
51. Character frequency.
52. First non-repeating character.
53. Anagram check.
54. Replace vowels with *.
55. Remove duplicate characters.
56. Capitalize each word.
57. Substring check.
58. ASCII values.
59. Count words.
60. Remove special characters.
61. Only digits check.
62. Longest word.
63. Rotate string left.
64. Rotate string right.
65. All substrings.
66. Pangram check.
67. Reverse words.
68. Count uppercase/lowercase.
69. Duplicate characters.
70. Sort characters.
71. Longest substring without repeating chars.
72. Longest palindrome substring.
73. String compression.
74. Decode compressed string.
75. All permutations.

76. Custom split().

77. Custom join().

78. Balanced parentheses.

79. Password validator.

80. Roman to integer.

81. Integer to Roman.

82. Smallest window containing chars.

83. Count overlapping substrings.

84. Minimum edits (basic idea).

C. Lists (Core Logic Training)

85. Largest element.

86. Second largest.

87. Smallest.

88. Remove duplicates.

89. Reverse list.

90. Sort without sort().

91. Bubble sort.

92. Selection sort.

93. Insertion sort.

94. Binary search.

95. Merge two lists.

96. Merge sorted lists.

97. Common elements.

98. Union of lists.

99. Intersection.

100. Frequency count.

101. List palindrome.

102. Sum of elements.

103. Multiply elements.

104. Even numbers list.

105. Odd numbers list.

106. Remove evens.

107. Remove odds.

108. Flatten nested list.

109. Rotate list by k.

110. Check sorted.

111. Find duplicates.

112. Pair with given sum.

113. Triplet with given sum.

114. Majority element.

115. Move zeros to end.

116. Missing element in sorted list.

117. Peak element.

118. Subarray with max sum.

119. Equilibrium index.

120. Kth largest.

121. Longest increasing subsequence (basic).

122. All subarrays.

123. Separate positive/negative.

124. Count primes in list.

125. Median.

126. Average.

127. Swap first and last.

128. Create squares list.

129. Create cubes list.

D. Tuples & Sets

130. List → tuple.

131. Tuple → list.

132. Max/min in tuple.

133. Count in tuple.

134. Remove duplicates using set.

135. Union of sets.

136. Intersection of sets.

137. Difference.

138. Symmetric difference.

139. Subset check.

140. Disjoint check.

141. Convert string to set.

142. Manual set implementation.

143. Find elements appearing once.

144. Convert list of tuples to dict.

145. Remove duplicate tuple entries.

E. Dictionaries

146. Student marks dictionary.

147. Access value.

148. Update value.

149. Delete key.

150. Merge dictionaries.

151. Character frequency.

152. Word frequency.

153. Invert dictionary.

154. Sort by keys.

155. Sort by values.

156. Sum values.

157. Key with max value.

158. Second highest value key.

159. Group words by length.

160. Group anagrams.

161. Nested dictionary.

162. Flatten nested dictionary.

163. Create dict from two lists.

164. LRU cache idea (basic).

165. Reverse dictionary safely.

F. Loops & Patterns

166. Print 1–100.

167. Even numbers.

168. Multiplication table.

169. Fibonacci.

170. Prime check.

171. All primes till n.

172. Sum of n numbers.

173. Power without **.

174. Armstrong till n.

175. Guessing game.

176. Password loop.

177. Triangle pattern.

178. Pyramid.

179. Inverted pyramid.

180. Diamond pattern.

181. Hollow square.

182. Pascal’s triangle.

183. Floyd’s triangle.

184. Multiplication matrix.

185. Number pyramid.

G. Functions & Recursion

186. Simple function.

187. Default arguments.

188. Keyword arguments.

189. Variable length args.

190. Recursive factorial.

191. Recursive Fibonacci.

192. Lambda example.

193. Map implementation.

194. Filter implementation.

195. Reduce implementation.

196. Custom max function.

197. Custom sort function.

198. Closure example.

199. Memoization Fibonacci.

200. Execution time decorator.

201. Authentication decorator.

202. Function factory.

203. Generator Fibonacci.

204. Flatten list recursion.

205. Strong password checker.

H. File Handling

206. Write file.

207. Read file.

208. Append file.

209. Count words.

210. Count lines.

211. Count characters.

212. Remove blank lines.

213. Replace word.

214. Word frequency in file.

215. Remove duplicate lines.

216. Compare two files.

217. Caesar cipher encrypt.

218. Decrypt file.

219. Merge files.

220. Split large file.

I. Exception Handling

221. ZeroDivision handling.

222. ValueError handling.

223. Custom exception.

224. Multiple except.

225. Finally block.

226. Raise exception.

227. Input validation loop.

228. FileNotFound handling.

229. Banking exception simulation.

230. Retry mechanism.

231. Safe division function.

232. Log errors to file.

233. Weak password exception.

234. Nested try-except.

235. Exception chaining.